

The GNS representation of a pure state is irreducible

`MathlibAnnex.Analysis.CStarAlgebra.isIrreducible_pureState_gnsStarAlgHom`

theorem

Applies the cyclic pure-state criterion to the space and vector constructed from the same state.

Statement

Let A be a unital complex C^* -algebra, and let φ be a pure state: a positive normalized continuous complex-linear functional extreme in the real state space. Let $(H_\varphi, \rho_\varphi, \xi_\varphi)$ be its GNS representation. Then the only closed reducing subspaces of H_φ for ρ_φ are 0 and H_φ .

Assumptions

The representation and space in the conclusion are the canonical GNS construction from φ , not an independently chosen representation on an arbitrary Hilbert space. No separability hypothesis is present.

Conclusion

The canonical GNS star-algebra homomorphism is irreducible. Its norm-one cyclic vector also ensures that its Hilbert space is nonzero.

Proof route

Verify the norm, cyclicity and pure vector-functional hypotheses of the cyclic irreducibility criterion.

▼ Proof steps

1. The GNS construction for the positive functional underlying φ supplies

$$\|\xi_\varphi\| = 1, \quad \overline{\{\rho_\varphi(a)\xi_\varphi : a \in A\}} = H_\varphi.$$

Normalization $\varphi(1) = 1$ gives the first equation; cyclicity of the construction gives the second.

2. For every $a \in A$, the GNS inner-product identity is

$$\langle \xi_\varphi, \rho_\varphi(a)\xi_\varphi \rangle = \varphi(a).$$

Hence the entire vector functional of ξ_φ equals the same pure state φ , rather than merely agreeing at the unit.

3. Apply Cyclic pure vector states force irreducibility with ρ_φ and ξ_φ . Step 1 verifies the norm-one and dense-orbit inputs; Step 2 transports the given purity to its vector-functional input. The output is exactly irreducibility of ρ_φ .

4. To see the mechanism of the criterion, let $K \subseteq H_\varphi$ be a closed reducing subspace, and let P be its orthogonal projection. Then $P^* = P = P^2$ and $P\rho_\varphi(a) = \rho_\varphi(a)P$ for every a . Define the positive functional

$$\psi(a) = \langle P\xi_\varphi, \rho_\varphi(a)P\xi_\varphi \rangle.$$

Orthogonality of K and $K^\perp$, and invariance of both, give

$$\varphi(a) = \psi(a) + \langle (I - P)\xi_\varphi, \rho_\varphi(a)(I - P)\xi_\varphi \rangle.$$

Hence $0 \leq \psi(a) \leq \varphi(a)$ for every positive a . Apply A positive functional dominated by a pure state to the given pure φ and this ψ : there is $t \in [0, 1]$ such that $\psi = t\varphi$.

5. For arbitrary $a, b \in A$, self-adjointness and idempotence of P , followed by its commutation with the representation, yield

$$\begin{aligned} \langle \rho_\varphi(a)\xi_\varphi, P\rho_\varphi(b)\xi_\varphi \rangle &= \langle P\rho_\varphi(a)\xi_\varphi, P\rho_\varphi(b)\xi_\varphi \rangle \\ &= \langle \rho_\varphi(a)P\xi_\varphi, \rho_\varphi(b)P\xi_\varphi \rangle \\ &= \psi(a^*b) = t\varphi(a^*b) \\ &= t\langle \rho_\varphi(a)\xi_\varphi, \rho_\varphi(b)\xi_\varphi \rangle. \end{aligned}$$

Here the inner product is linear in its second argument. For fixed b , density of the vectors $\rho_\varphi(a)\xi_\varphi$ makes $P\rho_\varphi(b)\xi_\varphi = t\rho_\varphi(b)\xi_\varphi$. Density in b and continuity now give $P = tI$ on all of H_φ .

6. Finally $P^2 = P$ implies $(t^2 - t)\xi_\varphi = 0$. The norm-one vector ξ_φ is nonzero, so $t^2 = t$ and $t \in \{0, 1\}$. Thus $P = 0$ or $P = I$, and its range K is respectively 0 or H_φ , as required.

Main citations

- Exact theorem and proof
- Cyclic pure vector states force irreducibility
- Purity in the real state space
- A positive functional dominated by a pure state

Lean source signature (exact)

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theorem isIrreducible_pureState_gnsStarAlgHom
  (phi : A →L[ℂ] ℂ) (hphi : phi ∈ stateSpace A) (hpure : IsPureState A phi) :
  StarAlgHom.IsIrreducible
    (positiveLinearMapOfMemStateSpace phi hphi).gnsStarAlgHom
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In the source

(phi : A →L[ℂ] ℂ)

Mathematical meaning

The given continuous complex-linear functional φ on the unital ordered C^* -algebra A .

In the source	Mathematical meaning
<code>(hphi : phi ∈ stateSpace A)</code>	φ is positive and $\varphi(1) = 1$; this permits its positive-map and GNS construction.
<code>(hpure : IsPureState A phi)</code>	The same φ is an extreme point of the real state space.
<code>positiveLinearMapOfMemStateSpace phi hphi</code>	The same functional, now carrying the positivity supplied by its state hypothesis.
<code>(positiveLinearMapOfMemStateSpace phi hphi).gnsStarAlgHom</code>	The canonical representation $\rho_\varphi : A \rightarrow \mathcal{B}(H_\varphi)$ on the GNS space of that functional.
<code>StarAlgHom.IsIrreducible</code>	For that represented action, every closed reducing complex subspace is either zero or the entire GNS space.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.isIrreducible_pureState_gnsStarAlgHom

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